What, Where, and How to Transfer in SAR Target Recognition Based on Deep CNNs

Zhongling Huang, Zongxu Pan, *Member, IEEE*, and Bin Lei

***Abstract*— Deep convolutional neural networks (DCNNs) have attracted much attention in remote sensing recently. Compared with the large-scale annotated data set in natural images, the lack of labeled data in remote sensing becomes an obstacle to train a deep network very well, especially in synthetic aperture radar (SAR) image interpretation. Transfer learning provides an effective way to solve this problem by borrowing knowledge from the source task to the target task. In optical remote sensing application, a prevalent mechanism is to fine-tune on an existing model pretrained with a large-scale natural image data set, such as ImageNet. However, this scheme does not achieve satisfactory performance for SAR applications because of the prominent discrepancy between SAR and optical images. In this article, we attempt to discuss three issues that are seldom studied before in detail: 1) what network and source tasks are better to transfer to SAR targets; 2) in which layer are transferred features more generic to SAR targets; and 3) how to transfer effectively to SAR targets recognition. Based on the analysis, a transitive transfer method via multisource data with domain adaptation is proposed in this article to decrease the discrepancy between the source data and SAR targets. Several experiments are conducted on OpenSARShip. The results indicate that the universal conclusions about transfer learning in natural images cannot be completely applied to SAR targets, and the analysis of what and where to transfer in SAR target recognition is helpful to decide how to transfer more effectively.**

***Index Terms*— Deep convolutional neural networks (DCNN), domain adaptation, synthetic aperture radar (SAR) target recognition, transfer learning.**

I. Introduction

DEEP learning techniques, which automatically learn effective hierarchical features from the large-scale data set, have been widely used in remote sensing data analysis in recent years. However, scarce labeled data, the biggest obstacle of applying deep learning to the field of remote sensing, still exist and significantly restrict further development. Different from tasks that have millions of labeled samples in natural image fields, the training data in the remote sensing field are usually inadequate to train a deep network well. Instead of training a deep network from scratch with a few data, transfer learning, which aims to transfer knowledge from the source domain with a large-scale data set to the target domain, provides an effective way to train a deep network with limited data. The most straightforward and commonly used trick is to fine-tune the network based on a pretrained one.

Remote sensing data, mainly from optical (multispectral and hyperspectral) and synthetic aperture radar (SAR) sensors, are multimodal with different imaging geometries and content. Penatti *et al*. [1] first indicated that the deep features can be generalized from everyday objects in daily images to objects in optical remote sensing images. Different kinds of convolution neural networks (CNNs), such as CaffeNet, AlexNet, and VGG, trained on ImageNet, a natural image data set, are tried to transfer to three bands' optical remote sensing image classification and achieve a remarkable performance [2]. Many subsequent research studies choose a variety of existing successfully pretrained CNN models on ImageNet to transfer to various tasks, such as image registration [3], airplane detection [4], scene classification [5], [6], image segmentation [7], and image superresolution [8], for both hyperspectral and multichannel remote sensing images. For optical remote sensing applications, transferring knowledge from natural images is prevalent since imaging mechanisms of both natural and optical remote sensing images are the same so that they can share some low- and mid-level features, such as those resemble either Gabor filters or color blobs. Apart from taking natural images as the source data, remote sensing data obtained from other platforms can also be used. Windrim *et al*. [9] transfer CNNs trained from certain hyperspectral images (HSI) to classify other HSI from different satellite(aerial) platforms. Similarly, Samat *et al*. [10] transfer between training and validation data of HSIs with domain adaptation to weaken the statistical distribution difference.

Due to different imaging mechanisms, approaches for the interpretation of optical remote sensing images cannot be directly used for interpreting SAR images in general. While transfer learning has begun to attract attention in optical remote sensing applications recently, relevant study in SAR images has not caught up with yet. We just find a few studies in which transfer learning is applied to conquer the difficulty of lacking labeled SAR data to train a deep network. Yang *et al*. [11] made the classifier learn the common knowledge among different target-aspect angles of SAR targets

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via transfer learning. Malmgren-Hansen *et al*. [12] proposed a generation approach on SAR data and answered the question about how to transfer the knowledge from the simulated SAR data to the real one. Huang *et al*. [13] indicated that features learned from a large amount of unlabeled SAR scene images via stacked convolutional autoencoders are transferable to a SAR target recognition task. To the best of our knowledge, there is not yet adequate evidence indicating whether the optical images can or cannot be transferred to SAR images with effect. Although several studies have attempted to explore how to transfer knowledge from optical to SAR images, even the rationality of this transfer is still under debate. Pros are as follows: Kang and He [14] utilized the intermediate layers of the pre-trained network on the CIFAR-10 data set as the feature extractor for classification of TerraSAR-X images and Wang *et al*. [15] fine-tuned the VGG-16 model trained under natural images to detect ships in SAR images. While cons also exist, for example, Marmanis *et al*. [16] thought that initialization with the weights learned from optical images has little effect on classification of SAR data, simply because the distributions of optical images and SAR data are probably too different from each other to transfer even in low layers.

Considering the particularity of SAR images, especially the different imaging mechanisms between optical and SAR sensors, it is not easy to transfer the features immediately from those successfully models that are often trained with the natural image data set [17]. The problem of transferring from other data sets to remote sensing data with large variations still remains to be solved and the transferability of trained networks to other imaging modalities needs to be further investigated [18]. In this article, we will explore transfer learning focusing on SAR target recognition in a further way and try to discover more properties.

The contribution of this article is to answer the following three questions about transfer learning via CNNs for SAR target recognition.

A. What to Transfer

The network and source tasks should be both considered in transfer learning on SAR target recognition. A deeper well-trained network with a large-scale data set generally has a stronger ability in extracting generic features, and the distance between source data and target data affects the transferability of features. We explore the influence of different source data and tasks, including optical images, SAR scene images, and SAR target data set, as well as classification and reconstruction, and different architectures to show what network together with data sets should be transferred to SAR target recognition. In addition, we propose a transitive transfer method via multisource data to improve the generality of features in layers significantly.

B. Where to Transfer

The transferability of features varies from layer to layer in deep CNNs (DCNNs). Some are general meaning that they are applicable to other tasks, while others are more specific to a particular task. Generally, the transferability of features decreases from low level to high level. We analyze the generality and specificity of features in different layers with various source data when taking SAR target recognition as the target task, so as to decide which level of the features can be used as the off-the-shelf representation for the target task.

C. How to Transfer

To transfer features that are specific to a particular task from the source task to the target task, we propose a method based on multikernel maximum mean discrepancy (MK-MMD) in domain adaptation, which combines the unsupervised and supervised learning to utilize the best of source and target data regardless of the labels. The approach increases the generality of features in task-specific layers, resulting in a stronger feature representation of the target data and a better performance in recognition.

D. SAR-Specific Model

We provide the SAR specific model pretrained on a large-scale SAR land cover and land use data set with a strong ability to extract spatial features of SAR images, which is validated to be well transferred to SAR targets, such as moving and stationary target acquisition and recognition (MSTAR) and OpenSARShip data sets [19].

The rest of this article is organized as follows. After a brief introduction of transfer learning and domain adaptation in Section II, the proposed method is detailed in Section III. Experiments and discussions are presented in Section IV to validate the effectiveness of the proposed method. Finally, the conclusion is drawn in Section V.

II. Related Work

In this article, we are interested in transfer learning on SAR target recognition. Consequently, we will introduce some typical research studies on SAR target recognition with transfer learning methods in this section first, and then followed by several related research studies about transfer learning and domain adaptation.

The simulated SAR images of vehicles with dense sampling of objects in different view angles are used for pretraining the CNN model to learn generic features that can be transferred to real SAR images in automatic target recognition (ATR) applications, proposed by Malmgren-Hansen *et al*. [12] for the first time. However, the simulation of SAR images requires high technology but the technique is not so mature to simulate enough reliable models and difficult to popularize. On the other hand, Huang *et al*. [13] found the features from unlabeled SAR scene images trained with stacked convolutional autoencoders are transferable to SAR targets. Although this is impressive and helpful under the case of lacking enough SAR targets but with adequate unlabeled SAR images, how generic or specific are the features from different source tasks transferred to SAR targets is still unknown. The transferability of features needs to be further explored and the discrepancy between source data and SAR targets should be fully taken into consideration to improve the performance of transfer learning.

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Transfer learning, usually aiming at transferring knowledge from a large data set known as the source domain to a small data set called the target domain [20], is widely popularized in DCNNs based approaches. Yosinski *et al*. [21] discussed the transferability of features in deep neural networks, taking AlexNet trained on ImageNet as an example. They proposed a method to analyze how transferable the features are and found the generalization of features to other data sets and tasks apparently decreases as the layer goes deeper, leading to more specific features to a particular data set or task, especially in layers 6 and 7, which is widely applied in the subsequent studies [22]–[24]. Although the coadaptation of neurons between layers will bring out the optimization difficulty, fine-tuning the transferring features on the target data set can disentangle this issue. Azizpour *et al*. [22] investigated several influencing factors on transferability, including network structure, early stopping, fine-tuning, similarity between source and target tasks, and so on. Among these factors, the similarity between source and target tasks is the most significant one to determine whether the learned representation is generic or not. Considering the large domain discrepancy, Tan *et al*. [25] proposed a transitive transfer learning method to transfer knowledge even when the source and target domains share few factors directly, with the aid of some annotated images as the intermediate domain to bridge them. Then they proposed a selective learning algorithm to transfer from face to airplane images that are totally different from each other [26]. This also inspires us to think whether the intermediate task closer to SAR target recognition is capable to increase the feature generality.

Domain adaptation approaches are often adopted to decrease the domain discrepancy between source and target tasks in transfer learning. MMD, a distance between embeddings of the probability distributions in a reproducing kernel Hilbert space (RKHS), proposed by Borgwardt *et al.* [27], is used as the discrepancy metric between the source and the target in domain adaptation and transfer learning [28]–[30]. Long *et al*. proposed the deep adaptation network [23], residual transfer networks [31], and joint adaptation network [32] successively based on the idea of domain adaptation using MK-MMD metric to reduce the domain discrepancy between the source and the target, which inspired us to focus on learning transferable features in SAR target recognition. In most domain adaptation problems, the labeled source data and unlabeled target data are used, with common or similar categories but different distributions, such as Office-31 data set that consists of 4652 images within 31 categories collected from different environment variations of Amazon (downloaded from amazon.com), Webcam (taken by web camera), and DSLR [taken by digital single-lens reflex (SLR) camera]. In our case, however, the categories of SAR targets to be recognized are usually never seen before so the classifier should be retrained. Moreover, those methods of transferring among natural images are probably not applicable to SAR targets. In this article, we will explore the specialized regulations and approaches specific to SAR target recognition.

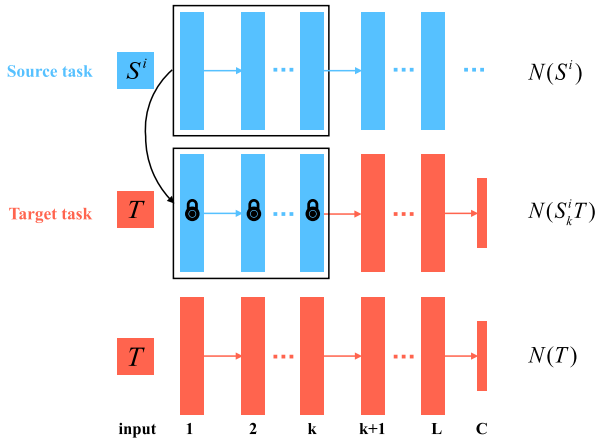


Fig. 1. Method of analyzing the generality of features. $N(S^i_kT)$ is obtained by transferring and fixing the first k layers of $N(S^i)$ then training the remaining randomly initialized layers on T. If the performance of $N(S^i_kT)$ is better than the performance of $N(T)$, the features in layer k of $N(S^i)$ is considered as generic, specific otherwise.

III. Methods

According to the three questions of what, where, and how to transfer in SAR target recognition that this article prepare to explore, we will first elaborate the method of analyzing the transferability of features and then propose our approaches to make full use of the transferred features.

A. Generic or Specific

The features extracted from different layers of deep convolutional networks can be grouped into two categories: the generic feature and the specific one. Features with generality means they are capable to represent other data set and those with specificity are closely related to the chosen data or tasks. In order to analyze the transferability of features in different layers in the SAR target recognition case, we adopt the method of qualifying the generality versus specificity of features in each layer of a DCNN [21]. Suppose there are n different source tasks to transfer. Denote the ith and jth source tasks as S^i and S^j, respectively, and the target task as T. For a network N with L layers, we would like to explore: 1) whether the features from the kth layer are generic to the target task or specific to the source task and 2) from which layer does the transferability of features decline dramatically.

First, we train the network N on source task S^i from scratch, denoted as $N(S^i)$. Then the network is trained on T, with the $1 \sim k$th layers copied from $N(S^i)$ and fixed as a feature extractor of the target task T and the $k+1 \sim L$th layers, as well as the classification layer C randomly initialized, as shown in Fig. 1. If the performance of this transferred network on T, denoted as $N(S^i_kT)$, is better than the performance of the retrained network on T, denoted as $N(T)$, the features in layer k are declared to be general. Otherwise, they are deemed to be specific to S^i. We compare the performance of $N(S^i_kT)$ and $N(S^j_kT)$ to evaluate the degree of generality of the kth

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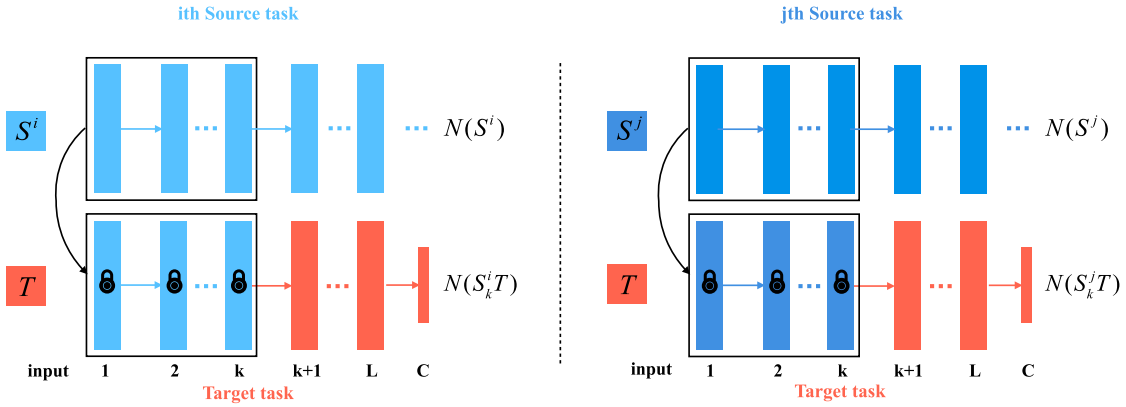


Fig. 2. To compare the transferability of different source tasks, different networks of $N(S^i_kT)$ and $N(S^j_kT)$ are obtained. If the performance of $N(S^i_kT)$ is better than the performance of $N(S^j_kT)$, S^i appears more suitable to extract general features in layer k than S^j.

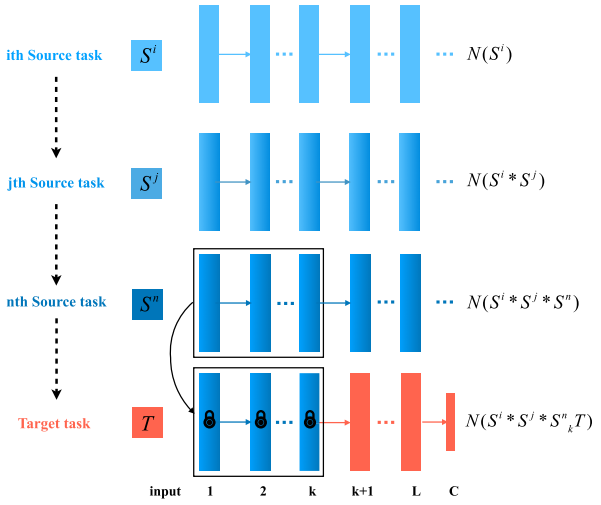


Fig. 3. Transitive transfer via multisource from S^i to S^n and the first k layers transferred to T at last.

layer features from different source tasks S^i and S^j, as shown in Fig. 2. The results are given in Section IV.

B. Transitive Transfer via Multisource

In this article, we propose the transitive transfer via multisource data sets. In the field of SAR image interpretation, various kinds of tasks, such as image classification, reconstruction, target detection, and recognition, are individually solved. Even for similar tasks, different problems usually do not cross paths with each other. Taking target recognition as an example, recognizing targets in optical images and SAR images, or recognizing different kinds of SAR targets such as airplanes and ships, is usually looked upon as different problems. Deep learning is a powerful tool to complete those tasks but training a new network for each task is time-consuming and data hungry for some tasks with limited labeled data. What if transitively transferring the knowledge task by task, especially from remotely similar tasks to similar ones? Can it be helpful to enhance and enrich the ability of feature extraction on target data set? In our method, as shown in Fig. 3, given a network $N(S^i)$ trained on S^i, we simply fine-tune all layers on S^j to fit the jth source task, obtaining the network $N(S^i * S^j)$. Similarly, we get $N(S^i * S^j \ldots * S^n)$ with knowledge from source data S^i, $S^j, \ldots$, and S^n. Then we will analyze the transferability of features in each layer similar to Section III-A. The results are given in Section IV.

C. Transfer Learning With Domain Adaptation

According to the previous analysis, specific features constrain transferring among various tasks. To solve this issue, we propose a transitive transfer-based method with domain adaptation to decrease the discrepancy between the source and target tasks. First, we will introduce the MK-MMD and then the two algorithms of the proposed method will be presented.

1) MK-MMD: MMD was first proposed by Borgwardt *et al.* [27] as the discrepancy metric to compare the distributions based on two sets of data. In transfer learning, most domain adaptation methods are based on the MMD to narrow the gap between source and target domains. Suppose the distributions of the source data $\{x^s\}$ and target data $\{x^t\}$ are p and q, respectively. For two data sets $\mathcal{D}_s = \{(x^s_i, y^s_i)\}^m_{i=1}$ and $\mathcal{D}_t = \{(x^t_j, y^t_j)\}^n_{j=1}$ with different distributions p and q, their MMD is defined as

$$\mathrm{MMD}(x^s, x^t) = \sup_{||\phi||_{\mathcal{H}} \leqslant 1} [E_{x^s \sim p}[\phi(x^s)] - E_{x^t \sim q}[\phi(x^t)]]_{\mathcal{H}} \tag{1}$$

where $\phi(\cdot)$ denotes an element of a set of functions in the unit ball of an RKHS $\mathcal{H}$ and $E_{x^s \sim p}[\cdot]$ denotes the expectation of x^s with the distribution p. In RKHS, the expectation is referred to as the embedding of p and denoted as μ_p for short, that is,

$$\mu_p = E_{x^s \sim p}[\phi(x^s)]. \tag{2}$$

As a result, the MMD can be regarded as a distance between embeddings of the probability distributions in an RKHS, which represents a metric of source and target data. Furthermore, the square of MMD can be written as

$$\begin{aligned} \mathrm{MMD}^2(x^s, x^t) = {} & E_{x^s \sim p}\langle \phi(x^s), \phi(x'^s) \rangle_{\mathcal{H}} \\ & + E_{x^t \sim q}\langle \phi(x^t), \phi(x'^t) \rangle_{\mathcal{H}} \\ & - 2E_{x^s \sim p, x^t \sim q}\langle \phi(x^s), \phi(x^t) \rangle_{\mathcal{H}} \end{aligned} \tag{3}$$

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where $\langle\cdot\rangle_{\mathcal{H}}$ denotes the inner product in RKHS $\mathcal{H}$ and the feature map $\phi(\cdot)$ can be associated with the kernel map $k(x^s, x^t) = \langle\phi(x^s), \phi(x^t)\rangle_{\mathcal{H}}$ in RKHS. Consequently, the empirical estimate of MMD can be given by

$$\text{MMD}^2(\mathcal{D}_s, \mathcal{D}_t) = \frac{1}{m^2}\sum_{i,j=1}^{m} k\left(x_i^s, x_j^s\right) + \frac{1}{n^2}\sum_{i,j=1}^{n} k\left(x_i^t, x_j^t\right) - \frac{2}{mn}\sum_{i,j=1}^{m,n} k\left(x_i^s, x_j^t\right). \quad (4)$$

The kernel k is usually defined as the convex combination of U basis kernels

$$k(x^s, x^t) = \sum_{u=1}^{U} \beta_u k_u(x^s, x^t), \quad \text{s.t. } \beta_u \geqslant 0, \sum_{u=1}^{U}\beta_u = 1 \quad (5)$$

and in our method, the Gaussian kernel function is selected as the basis kernel.

In order to use minibatch stochastic gradient descent (SGD) more easily and less time-consumingly in CNN, Gretton *et al.* [33] proposed the unbiased estimate of MK-MMD with linear complexity, which gives an approximation of a summation form. Given a quad-tuple $z_i = (x_{2i-1}^s, x_{2i}^s, x_{2i-1}^t, x_{2i}^t)$, by supposing $m = n$, the square of MMD can be rewritten as

$$\text{MMD}^2(\mathcal{D}_s, \mathcal{D}_t) = \frac{2}{m}\sum_{i=1}^{m/2} h(z_i) \quad (6)$$

where

$$h(z_i) = k\left(x_{2i-1}^s, x_{2i}^s\right) + k\left(x_{2i-1}^t, x_{2i}^t\right) - k\left(x_{2i-1}^s, x_{2i}^t\right) - k\left(x_{2i}^s, x_{2i-1}^t\right). \quad (7)$$

2) Deep Domain Adaptation Based on Transitive Transfer With Multisource: In our method, we will choose a variety of source tasks with diverse similarity to the target task to assist in recognizing some new types of SAR targets by transitive transfer learning from distant to similar. Given a set of source tasks and arrange them in ascending order according to the similarity with the target task $S^1, S^2, \ldots, S^n$. We pretrain and fine-tune the network as proposed in Section III-B and analyze the transferability of features in each layer to see where the generality drops fiercely. Suppose the first k layers have the strong ability to extract general features of target data, denoted as the off-the-shelf layers, and the $k+1 \sim L$ layers are more specific than the previous layers, denoted as the adaptation layers. Since the $N(S^1 * \ldots * S^n)$ is fine-tuned on S^n at last, we only adapt the data sets of S^n and T.

In the popular domain adaptation methods [23], [32], [34], [35], the source data and target data share the same set of categories but with different probability distributions with the target data all unlabeled. The classification loss of source data and MMD between source and target data is combined to backpropagate to decrease the discrepancy, and then the target data can be classified into categories directly. In our case, however, the types of SAR targets to be recognized are never seen before and the classification layer should be retrained. In this article, we proposed two algorithms and will have an elaborate discussion in Section IV on how to choose an appropriate algorithm in different scenarios.

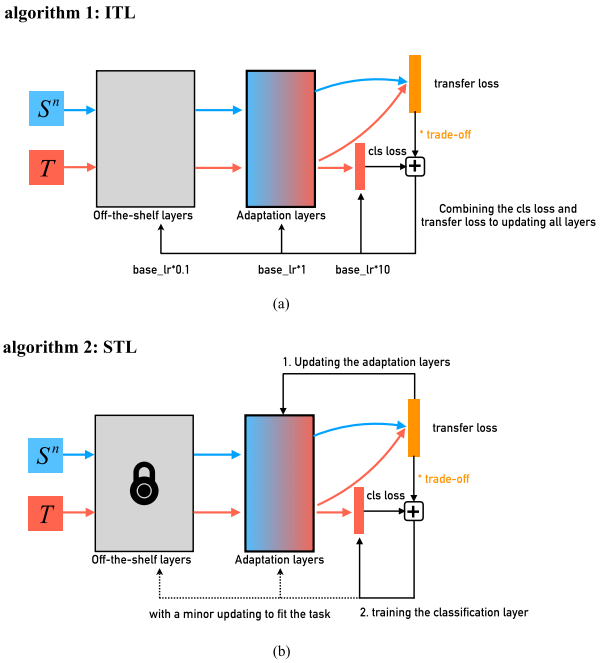


Fig. 4. ITL and STL training algorithms based on domain adaptation. (a) ITL algorithm that integrates the transfer loss in adaptation layers and the classification loss to backpropagate. (b) STL algorithm to update the adaptation layers with transfer loss only and then train the classification layer and slightly fine-tune the off-the-shelf layers.

First, an integrated learning algorithm that combines the classification and domain adaptation is proposed as integrated transfer learning (ITL), as shown in Fig. 4(a). Given a minibatch of a quad-tuple of $\{x_{2i-1}^s, x_{2i}^s, x_{2i-1}^t, x_{2i}^t\}$ as the input to the network, the transfer loss in the lth adaptation layer is calculated by (6) and (7), denoted as

$$\text{mmd}_l\left(x_{2i-1}^s, x_{2i}^s, x_{2i-1}^t, x_{2i}^t\right) \quad (8)$$

where $l = k+1, \ldots, L$. The classification loss of target data is calculated by the standard Softmax loss, denoted as $\mathcal{L}_C(x_{2i-1}^t, y_{2i-1}^t; x_{2i}^t, y_{2i}^t; \theta_C)$, where θ_C represents the category classifier. The network is trained by minimize the total loss of

$$\mathcal{L}_C\left(x_{2i-1}^t, y_{2i-1}^t; x_{2i}^t, y_{2i}^t; \theta_C\right) + \lambda\sum_{l=k+1}^{L}\alpha_l \text{mmd}_l\left(x_{2i-1}^s, x_{2i}^s, x_{2i-1}^t, x_{2i}^t\right) \quad (9)$$

where λ denotes the tradeoff between transfer loss and classification loss and α_l denotes the weight of transfer loss in each adaptation layer. In the ITL algorithm, the transfer loss in adaptation layers is only added as a regularizer to classification and λ is a dynamic parameter in the training process to keep a good balance on transfer loss and classification loss, especially at the later stage in training, λ should be reduced by 0.1 to get a better tradeoff. The setting of α depends on the transferability of each adaptation layer. Generally, the learning rate of the off-the-shelf layers should be smaller and the classification layer larger than the adaptation layers.

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TABLE I
MSTAR DATA SET

Category	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZSU23	Total
17°	299	233	298	256	233	299	299	232	299	299	**2747**
15°	274	195	274	295	196	274	273	196	274	274	**2425**

Second, considering the transfer loss and the classification loss are mutually interactive and restrictive when combined to optimize the parameters of the network, we propose a two-step training algorithm, namely, separated transfer learning (STL), as shown in Fig. 4(b). In the first step of training the adaptation layers, the off-the-shelf layers are frozen because of the generality of representing the target data, which also lowers the computational cost of optimizing the parameters. The transfer loss calculated by (8) is used to train the adaptation layers, aiming at decreasing the feature discrepancy in specific layers. Then the classification loss combined with the transfer loss is minimized to train the classification layer, with a minor updating in the off-the-shelf and adaptation layers. In the second step, λ is reduced by 0.1 than the first step to make the transfer loss play a subordinate role as a constraint term.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Data Sets and Tasks Description

In our experiments, we analyze the transferability of features using different source tasks and networks and evaluate the proposed method on the target task, OpenSARShip recognition. The alternative source data sets/tasks contain ImageNet, TerraSAR-X images, and SAR targets of MSTAR, for classification or reconstruction. In the following, the brief descriptions of these data sets and tasks are given.

1) ImageNet for Classification: ImageNet is a well-known large-scale data set of natural images in computer vision, providing the most comprehensive and diverse coverage of the image world [36]. It contains 3.2 million labeled images over 5247 categories, over 600 images for each category on average. Generally, a subset of the large hand-labeled ImageNet data set with 1.2 million images in 1000 object classes is considered as the benchmark to train the deep networks in the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) [37], where the remarkable DCNNs for image classification, such as AlexNet [38], GoogLeNet [39] and ResNet [40], are proposed in 2012, 2015, and 2016, respectively.

2) TerraSAR-X Images for Classification and Reconstruction: First, we collect over 50 000 SAR image slices without annotation. These SAR slices are randomly cropped from SAR scene images covering various landscapes from TerraSAR-X, a German Earth-observation satellite that provides high-quality and precise earth observation data of 3-m resolution with the StripMap mode. With rich texture information in those unlabeled SAR slices, a deep-stacked convolutional autoencoder is trained to reconstruct the slices, generating a series of hierarchical convolution layers capable of extracting efficient features.

In addition, a high-resolution SAR land-cover annotated data set [41], collected from TerraSAR-X horizontally polarized (HH), multilook ground range-detected (MGD) products, is applied for SAR land cover classification task in our experiments. The selected SAR images were taken in the high-resolution spotlight mode with the pixel spacing of 1.25 m, acquired with an incidence angle between 20° and 50°, and with descending and ascending pass directions. Covering 288 full scenes of urban and nonurban areas all over the world, such as cities in Africa, Asia, Europe, and some ocean areas, this data set is hierarchically annotated of 3 levels, 150 categories, and more than 100 000 patches. In our experiments, seven categories with a high-level annotation of *settlements, public transportation, industrial areas, agricultural land, natural vegetation, bare ground, and water bodies* are applied for classification, as shown in Fig. 5.

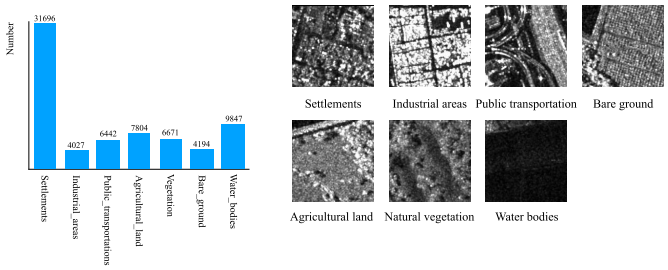


Fig. 5. Annotated TerraSAR-X land cover data set with seven categories out of nine in level 1.

3) MSTAR for SAR Target Recognition: The MSTAR public release data set [42] collected by the Sandia National Laboratory SAR sensor platform contains ten categories of military vehicles: the T72, BTR70, BMP2, 2S1, BRDM2, BTR60, D7, T62, ZIL131, and ZSU23, with the resolution of 1 ft on X-band. Those targets chips acquired at a depression angle of 17° are usually used as the training data, and 15° as the testing data to evaluate the SAR target recognition algorithms. Details of the MSTAR data set for ten-category SAR target recognition are given in Table I.

4) OpenSARShip for SAR Target Recognition: Huang *et al.* [43] present a SAR ship data set of Sentinel-1, containing 11 346 ship chips from 41 Sentinel-1 SAR images. The data set provides the single-look complex (SLC) and the ground range-detected (GRD) products of the interferometric wide swath (IW) mode, with polarization of VH and VV, as well as four formats of amplitude values, visualized data in gray scale, visualized data in pseudocolor, and radiometric calibrated data. OpenSARShip contains 17 types of ships, such as cargo, tanker, passenger, tug, and so on, but unbalanced numbers in each type (8470 in cargo and 4 in towing for example). There are five elaborated types in cargo, namely, cargo, container ship, bulk carrier, general cargo, and other cargo. In order to

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TABLE II
OPENSARSHIP DATA SET IN OUR EXPERIMENTS

Elaborated Type	Cargo	Bulk Carrier	Container Ship	Total
train	100	100	100	**300**
test	79	132	135	**346**

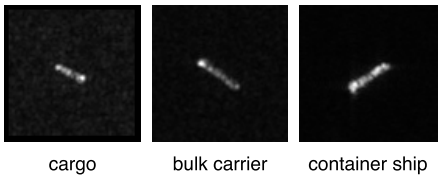


Fig. 6. Three elaborated types of OpenSARShip.

evaluate the method with limited target data and balance the training numbers of each category, we select the elaborated types of cargo, container ship, and bulk carrier of the GRD mode (with a resolution of 10 m) and VV polarization in our experiments, filtering those ship chips with the size larger than 70×70 pixel to ensure the sufficient image information. The details are given in Table II and shown in Fig. 6.

B. What to Transfer

In this section, we will discuss how the different networks and source tasks affect the transferability of features in SAR target recognition and then apply the conclusion to our subsequent experiments. Azizpour *et al.* [22] indicated that overparameterizing the network by increasing the width (the number of parameters at each layer) and depth (the number of convolution layers) can improve the performance on other data sets in transfer learning when they are close to the source tasks but may harm the transferability of features to distant target tasks. According to the previous studies, it is important to select the appropriate network and the source task to transfer in a SAR target recognition problem.

Aiming at recognizing the SAR ship targets of the OpenSARShip data set with only hundreds of labeled images, the first thought would be transferring layers from a close data set such as MSTAR. We will discuss different networks pretrained on the MSTAR data set in IV-B1. In addition, we will explore how other source data or tasks, such as ImageNet classification, SAR images reconstruction, and SAR land cover classification, perform on transferring to SAR target recognition in IV-B2, as well as the transitive transfer method using multisource tasks.

1) What Network: Three networks, A_ConvNet [44] that has the state-of-the-art performance on MSTAR recognition, H_Net [13] that is also well performed on MSTAR using the stacked convolutional autoencoders to learn hierarchical layers with unlabeled SAR images and transfer to SAR targets, and AlexNet [38] that is a breakthrough in the large-scale image classification with a deep neural network, are explored in this section. With more than 90% of parameters in fully connected layers, we only use the convolution layers of AlexNet due to the data scale in SAR targets, denoted as AlexNet_Conv.

TABLE III
CONFIGURATION OF CONVOLUTION LAYERS IN THREE NETWORKS

Network	A_ConvNet	H_Net	AlexNet_Conv
conv1	16(5)	48(5)	96(11)
conv2	32(5)	96(5)	256(5)
conv3	64(5)	128(3)	384(3)
conv4	128(6)	128(3)	384(3)
conv5	None	256(3)	256(3)
size	**0.4 M**	**0.7 M**	**4 M**

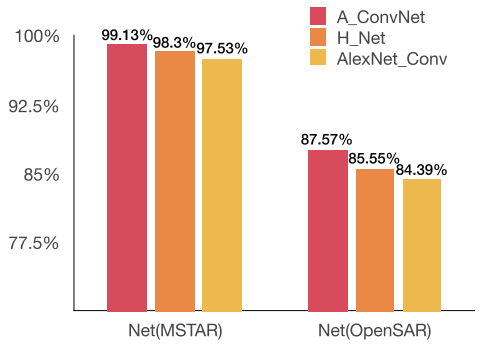


Fig. 7. Target recognition accuracy on MSTAR and OpenSARShip of A_ConvNet, H_Net and AlexNet_Conv, which are all trained from scratch.

As depicted in Table III, $a(b)$ denotes a channels and the kernel size of $b \times b$ in each convolution layer. We denote the network Net retrained on MSTAR and OpenSARShip as $Net(M)$ and $Net(O)$, respectively. It can be seen in Fig. 7 that as the network going deeper and wider from A_ConvNet to AlexNet_Conv, the performance on SAR targets is decreasing. A_ConvNet is successful in MSTAR because the smaller network offers an appropriate feature space to fit the limited data. When it comes to a deeper and wider network, a more complex and nonlinear function is going to be learned with limited data, which is difficult to find the optimal solution.

Next, we follow the instruction in Section III-A to analyze the feature transferability in each layer of the three models: $A_ConvNet(M)$, $H_Net(M)$, and $AlexNet_Conv(M)$. Considering the hyperparameter of the conv5 layer in A_ConvNet is specific to classification, only the first four convolution layers are transferred in our experiments. We record the recognition rate on OpenSARShip test data as $Net(M_k O)$, denoting the model trained by transferring and freezing the first k layers of the model $Net(M)$, where Net is in $\{A_ConvNet, H_Net, AlexNet_Conv\}$. The remaining higher layers together with the classification layer are randomly initialized and trained on OpenSARShip. The performance of $Net(M_k O)$ is shown in Fig. 8 and Table IV. Although A_ConvNet performs better on small-scale data sets like OpenSARShip and MSTAR training from scratch than H_Net and AlexNet_Conv, the features in each layer of $A_ConvNet(M)$ reflect low generality to OpenSARShip, observing that the performance of $A_ConvNet(M_k O)$ is not as good as $A_ConvNet(O)$. On the other hand, the overparameterized networks $H_Net(M)$ and $AlexNet_Conv(M)$ improve the performance on OpenSARShip in transfer learning.

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TABLE IV
OPENSARSHIP RECOGNITION RATE OF DIFFERENT NETWORKS AND SOURCE TASKS WHEN TRANSFERRING AND FREEZING DIFFERENT LAYERS

Network	Source Task	Net(OpenSAR)	Transferred Layers (Frozen)				
			1	2	3	4	5
A_ConvNet	MSTAR	0.8757	0.8612	0.8612	0.8670	0.8208	none
H_Net	SAR(recon)	0.8555	0.8641	0.8483	0.8223	0.7818	0.68
	MSTAR		0.8805	0.8728	0.8526	0.8410	0.8324
	SAR(recon)*MSTAR		**0.8844**	**0.88**	**0.8858**	**0.8902**	**0.8584**
AlexNet_Conv	ImageNet	0.8439	0.8901	0.8584	0.8468	0.8584	0.7774
	SAR		0.8974	0.8988	**0.8883**	**0.8921**	**0.8526**
	MSTAR		**0.9017**	**0.9075**	0.8859	0.8757	0.8511
AlexNet_Conv (transitive transfer)	ImageNet*SAR	0.8439	0.8930	0.8901	0.8718	0.8671	0.7109
	ImageNet*MSTAR		0.8931	0.8902	0.8815	0.8902	0.7283
	ImageNet*SAR*MSTAR		**0.8988**	0.8931	**0.8960**	**0.9017**	0.7486
	SAR*MSTAR		**0.8988**	**0.9032**	**0.8959**	0.8872	**0.8612**

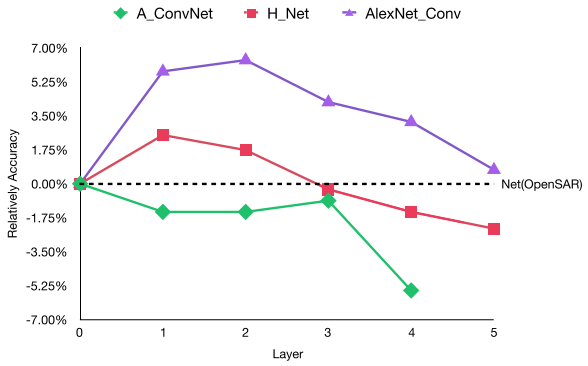


Fig. 8. Relative accuracy of $Net(M_k O)$, where Net is from $\{A_ConvNet, H_Net, AlexNet_Conv\}$ and k denotes the layer of $Net(\cdot)$.

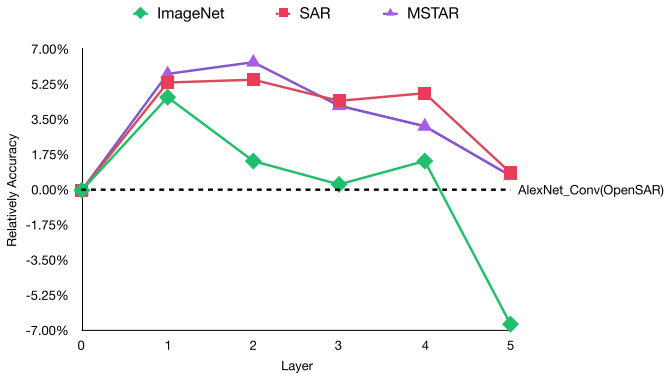


Fig. 9. Relative accuracy of $AlexNet_Conv(I_k O)$, $AlexNet_Conv(S_k O)$, and $AlexNet_Conv(M_k O)$, respectively, compared with the performance of $AlexNet_Conv(O)$, where k denotes the layer of $AlexNet_Conv(\cdot)$. The points above the black baseline indicate the generality of the features in the kth layer and those below the black baseline indicate the specificity.

It can be inferred that even though the difficulty for a small data set to find an optimal solution in training a deeper and wider network, the learned features are more general to a related task so that the transferring features are able to help the related target task find a better solution.

2) What Source Data/Tasks: Intuitively, we can imagine that the closer data or tasks are better to provide transferable features to SAR target recognition, such as other kinds of SAR target recognition and SAR land cover classification. However, in some cases, we do not have enough labeled SAR data to pretrain a deeper network with strongly representative features. On the other hand, the abundant unlabeled SAR images can be easily collected. Huang *et al.* [13] indicated that the large scale of unlabeled SAR scene data can be reconstructed with training a stacked convolution autoencoders of which the stacked convolutional layers are capable to transfer to SAR target recognition task. Still, whether the well-known natural images pretrained models popular in transferring to other remote sensing tasks are transferable to SAR targets remains to be explored.

First, we experiment the AlexNet_Conv pretrained with ImageNet denoted as $AlexNet_Conv(I)$, SAR land cover data set denoted as $AlexNet_Conv(S)$, and MSTAR denoted as $AlexNet_Conv(M)$, respectively, in transferring to OpenSARShip, and the results can be found in Fig. 9 and row 4 of Table IV. Compared with the source tasks of SAR land cover classification and MSTAR target recognition, the features in the first layer of $AlexNet_Conv(I)$ perform well in generalizing but show much specificity in higher layers, performing a significant drop when transferring and freezing the second to fifth convolution layers. Even though the low-level features learned from natural images that resemble Gabor filters are effective to represent SAR targets, the features from higher layers are more specific on natural images, which indicate that more distant the midlevel features of natural images and SAR targets present, much worse in high layers. On the other hand, features in $AlexNet_Conv(S)$ and $AlexNet_Conv(M)$ show more robust on generalization to SAR targets. More specifically, the SAR land cover classification-trained model performs better in higher layers due to the large-scale data set with abundant SAR image information and the similar task of classification.

What if we do not have the large-scale annotated SAR images to pre-train a deep network? Row 3 of Table IV shows how the unlabeled SAR images perform in transferring. Due to the distance between unlabeled SAR images' reconstruction task and the OpenSARship recognition, the transferability of features in $H_Net(S)$ decreases to be specific just in layer 2, while the MSTAR recognition task much more similar with our target task results in more general features in $H_Net(M)$.

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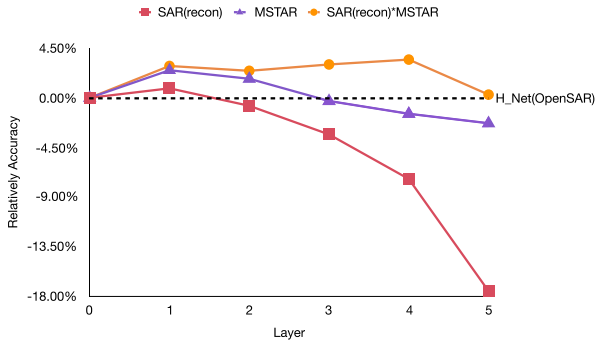


Fig. 10. Relative accuracy of $H_Net(S_k O)$, $H_Net(M_k O)$, and $H_Net(S{*}M_k O)$, respectively, compared with the performance of $H_Net(O)$, where k denotes the layer of $H_Net(\cdot)$. The points above the black baseline indicate the generality of the features in the kth layer and those below the black baseline indicate the specificity.

With limited SAR-annotated data in reality, it is not easy to find a source task that is similar to SAR targets and with a large amount of related data. With features specific to natural images in higher layers of ImageNet pretrained models and specific to reconstruction tasks of unlabeled SAR pretrained models, we are going to explore the transitive transfer method with multisource tasks related to SAR targets to enhance the generality of features in deep networks.

$H_Net(S * M)$ denotes the network of simply fine-tuning the convolution layers on $H_Net(S)$ with the MSTAR data set. Fig. 10 shows the performance of transferring different layers of the pretrained network to OpenSARShip recognition, comparing with the black line that denotes the performance of $H_Net(O)$. The areas above the black line indicate that the features are general to the target task and those below the line indicate the specificity. Strikingly, $H_Net(S * M)$ distinctly increases the generality of features in mid and high layers, which indicates although the distant source task of unlabeled SAR images reconstruction, the intermediate task of MSTAR classification has an impact on enhancing the transferability of features to other SAR target recognition tasks, on the condition that the pretrained model on SAR images reconstruction provides a good basis.

Now that the multisource transitive transferring performs well on feature generalization, we attempt to explore the ImageNet pretrained model transferring to SAR target recognition. Yosinski *et al.* [21] pointed out that the fragile coadaptation would affect the performances when freezing the first several layers. Our experiments prove that the effect of the coadaptation in training AlexNet_Conv with OpenSARShip can be ignored due to the tiny fluctuation, as shown in Fig. 11. $AlexNet_Conv(I)$ is fine-tuned with a subset of annotated SAR land cover data set with 12 000 slices for classification, obtaining $AlexNet_Conv(I * S)$. Similarly, $AlexNet_Conv(I * M)$ and $AlexNet_Conv(I * S * M)$ are obtained with the MSTAR data set. As shown in Fig. 12, the transferability of features from layer 2 to layer 4 is remarkably increased from $AlexNet_Conv(I * S)$ to $AlexNet_Conv(I * M)$, especially in $AlexNet_Conv(I * S * M)$, where the features generality of the fourth layer is comparable to the lower level features.

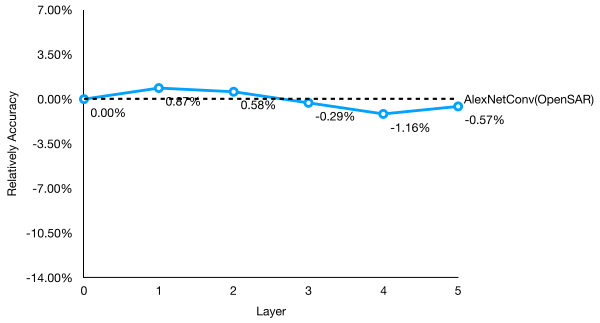


Fig. 11. Tiny fluctuation of freezing the first k layers of $AlexNet_Conv(O)$ and training the remaining layers randomly initialized with OpenSARShip data set. We will ignore the minor influence of coadaptation in our analysis.

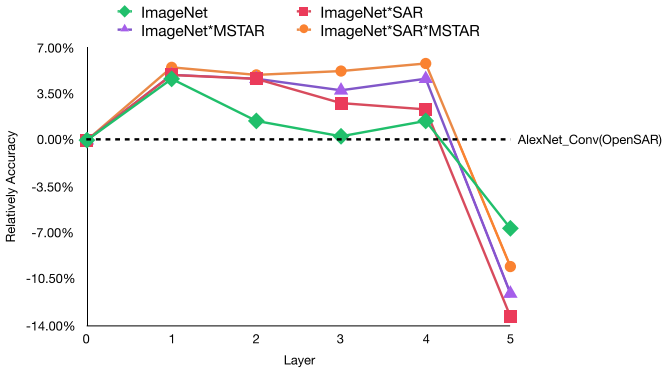


Fig. 12. Performances on different source tasks of AlexNet_Conv. The relative accuracy of $AlexNet_Conv(I_k O)$, $AlexNet_Conv(I * S_k O)$, $AlexNet_Conv(I * M_k O)$, and $AlexNet_Conv(I * S * M_k O)$, respectively, compared with the performance of $AlexNet_Conv(O)$, where k denotes the layer of $AlexNet_Conv(\cdot)$. The points above the black baseline indicate the generality of the features in the kth layer and those below the black baseline indicate the specificity.

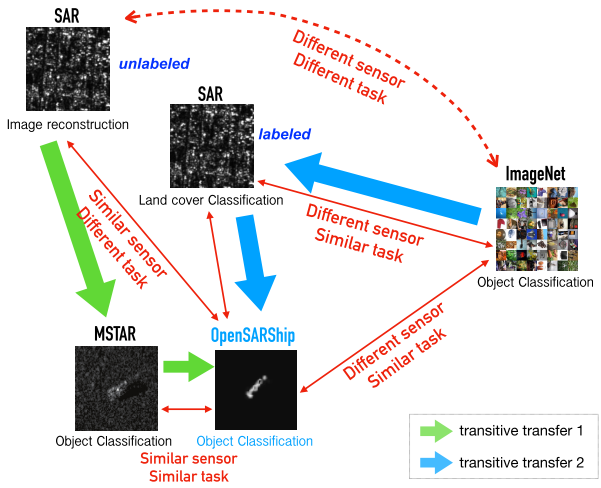


Fig. 13. Relationship between the target task (OpenSARShip recognition) and different source tasks (ImageNet classification, SAR scene image reconstruction, SAR scene image classification, and MSTAR target recognition).

The distance among those tasks is illustrated in Fig. 13. The abundant SAR scene images from similar sensors to SAR targets are suitable for pretraining a deep network to transfer to other SAR-related tasks with limited data, but the image reconstruction task with unlabeled data is distant to the SAR target recognition task that affects the transferability

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of features in mid and high layers. As a comparison, the MSTAR classification is close to OpenSARShip recognition, while the MSTAR data is limited to train a deeper and wider network. Consequently, if it is possible to obtain a large-scale annotated SAR image data set, the pretrained model will be very useful for SAR target recognition. If not, the unlabeled SAR images also help as transitive transfer 1 shown in Fig. 13, where the MSTAR classification task can build a bridge between unlabeled SAR image reconstruction and OpenSARShip recognition to improve the generality of features in layers. On the other way, if you want to take the use of the natural images pretrained models to SAR-related problems, we will give advice to learn some information from SAR images based on the model as transitive transfer 2 shown in Fig. 13, which is useful in decreasing the specificity to natural images of features in higher layers.

The analysis in this section reveals that the transferability of features is influenced by the generality of the transferred network and the distant between the source and target tasks, that is to say, the network and the source tasks both have an impact on transferring to the SAR target recognition task. Multisource transitive transferring is a good idea to combine different source data sets from large scale to limited, as well as from distant to similar, to obtain more general features. The network gradually learns more useful knowledge in the process of completing different tasks. Despite the large diversity between natural images and SAR targets, the low-level features are general and transferable, and fixing with more knowledge of SAR images via multisource transitive transferring can notably increase the transferability. Multisource transitive transferring method can not only adopt a larger network but also combine the greatly generic low-level features of training on ImageNet and the improving transferable features in higher layers.

C. Where and How to Transfer Effectively

In the previous research studies, Yosinski *et al.* [21] found that the performance drops in fully connected layers due to the representation specificity when transferring to other natural images. As a result, the follow-up studies [23], [34], [45] are accustomed to adapting features in each of the fully connected layers when transferring to other natural images. Moreover, Hu *et al.* [2], Zhao *et al.* [6], and Marmanis *et al.* [5] individually transfer the high-level features from the first fully connected layer of AlexNet to remote sensing images classification task. However, in our previous discussion, we find that this conclusion cannot be simply applied to SAR target recognition. In this section, we will discuss where to transfer features in different situations and how to transfer more effectively to reduce the discrepancy between the source and target domains.

The features in $AlexNet_Conv(S)$ are good enough to transfer to the SAR targets. For the MSTAR data set, the model achieves an overall accuracy of 99.34% by fine-tuning all layers, better than the state-of-the-art. For OpenSARShip, it performs with a fine-tuning result of 91.04%, which is 1.73%, 1.1%, 4.23%, and 2.6% better than $AlexNet_Conv(I*M)$, $AlexNet_Conv(I*S*M)$, $H_Net(M)$, and $H_Net(S*M)$, respectively. In this part, we mainly focus on the four

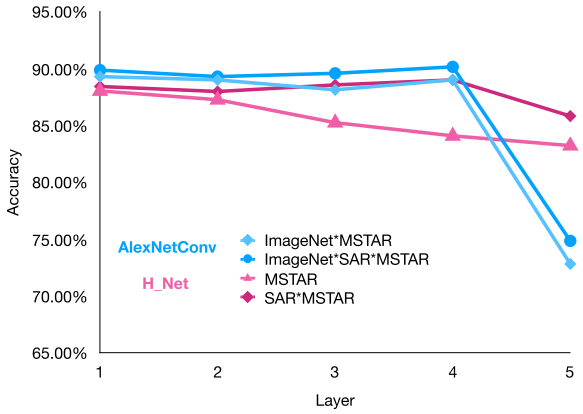


Fig. 14. Performance of using different source tasks to transfer to the OpenSARShip recognition task. The red series lines denote the H_Net and the blue series lines denote the AlexNet_Conv.

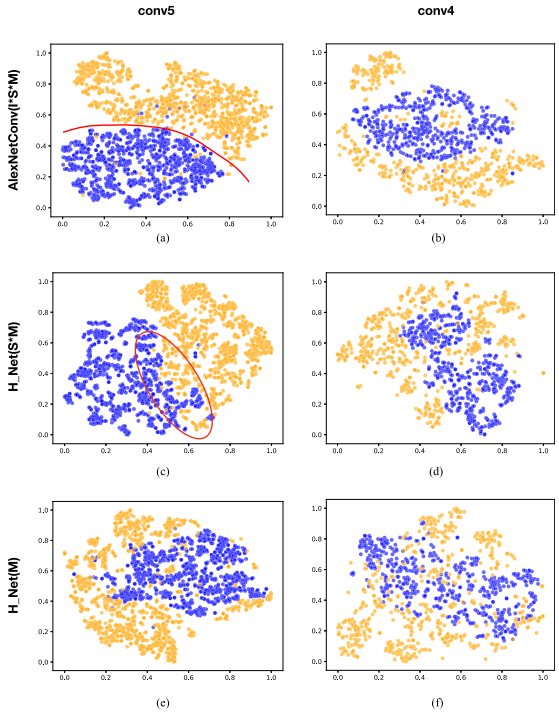


Fig. 15. Feature visualization of MSTAR and OpenSARShip data sets in layer 5 and layer 4 in different scenarios by t-sne. The blue dots denote the MSTAR and the orange dots denote the OpenSARShip. (a) and (b), (c) and (d), and (e) and (f) Network transferred from $AlexNet_Conv(I*S*M)$, $H_Net(S*M)$, and $H_Net(M)$, respectively. (a), (c), and (e) and (b), (d), and (f) Features from layer 5 and layer 4, respectively.

pretrained models from AlexNet_Conv and H_Net to see how to make them more effective in transferring to SAR targets. Fig. 14 presents the transferability of each layer in different scenarios. In $AlexNet_Conv(I*S*M)$ and $AlexNet_Conv(I*M)$ scenarios, the first four convolution layers show a strong ability to extract the generic features of OpenSAR but rapidly decreased in layer 5. However, in $H_Net(M)$ and $H_Net(S*M)$ scenarios, even though the performance of transferring the first four layers is not as good as AlexNet_Conv ones, the features in layer 5 present a better generalization. We visualize the features in layers 4 and 5 from MSTAR and OpenSARShip data sets of different scenarios by t-sne [46], as shown in Fig. 15, where the blue dots

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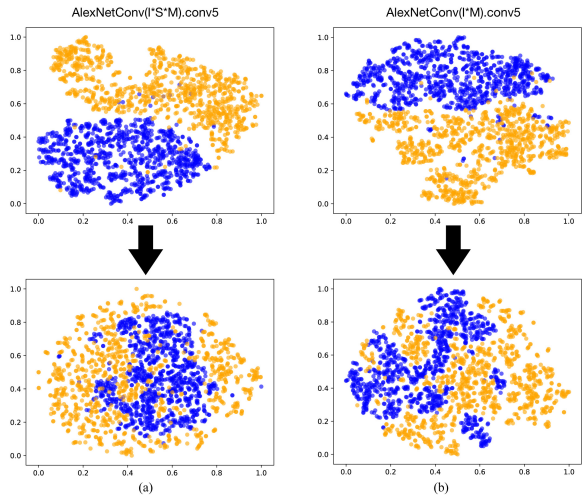


Fig. 16. Feature distribution in layer 5 becomes more similar after updating the adaptation layers. (a) $AlexNet_Conv(I*S*M)$ scenario. (b) $AlexNet_Conv(I*M)$ scenario.

denote the MSTAR data set and the orange ones denote the OpenSARShip data set. The features of MSTAR and OpenSARShip of layer 5 in $AlexNet_Conv(I*S*M)$ can be simply distinguished, which indicates the large difference of feature distributions between source and target data. In $H_Net(S*M)$, however, the feature distribution presents more indistinguishable between source and target data, more noticeable in $H_Net(M)$. These properties concern the choice of strategies of how to transfer the features in SAR target recognition.

We experiment with the ITL and STL algorithms proposed in Section III-C in different scenarios of AlexNet_Conv and H_Net.

1) AlexNet_Conv: For the STL algorithm, according to the previous analysis, we consider layer $1 \sim 4$ as the off-the-shelf layers in $AlexNet_Conv(I*M)$ and $AlexNet_Conv(I*S*M)$ because of the great performance of $AlexNet_Conv(I*M_4O)$ and $AlexNet_Conv(I*S*M_4O)$, while layer 5 as the adaptation layer. After the first step of updating the adaptation layer, we can observe an obvious improvement on similar feature distributions of source and target data, which implies the discrepancy of features in source and target is decreased, as shown in Fig. 16. In this step, the off-the-shelf layers are fixed since the quality of generality makes them possible to extract the off-the-shelf features of OpenSARShip data. Moreover, it is an unsupervised learning part so that all labeled and unlabeled OpenSARShip data can be used to narrow the gap of feature distributions. Next, the classification layer is trained with labeled OpenSAR data by combining the cross-entropy loss of labels and outputs of the Softmax layer and the transfer loss. In this part, the learning rate in layers 1–5 is set to 10^{-4} so that the previous layers are slightly fine-tuned and the learning rate of the classification layer is set to 10^{-2}, which is 100 times larger than the previous layers. The transfer loss constrains the whole network to maintain the property of narrowing the discrepancy between the source and the target, and it should be controlled by the tradeoff λ to avoid dominating the total loss and preventing the continuous decreasing of classification loss. In our experiments, we set λ as 1.5 in AlexNet_Conv scenarios.

TABLE V

PERFORMANCE OF DIFFERENT SCENARIOS BY USING ITL AND STL, COMPARED WITH FINE-TUNING ALL LAYERS

Network	AlexNet_Conv		H_Net	
Source Tasks	$I*M$	$I*S*M$	M	S(recon)$*M$
Fine-tune	89.31%	89.94%	86.41%	88.44%
ITL	89.88%	90.46%	87.28%	**89.88%**
STL	**90.75%**	**91.9%**	**88.43%**	89.31%

Table V gives the performance of different algorithms in different scenarios. STL approach boosts the performance by 1.44% and 1.96%, respectively, on $AlexNet_Conv(I*M)$ and $AlexNet_Conv(I*S*M)$ compared with the common fine-tuning methods on transfer learning. For the ITL approach of combining the transfer loss and the classification loss to fine-tune all layers, the results are not as good as the STL but still better than simply fine-tuning method, improving 0.57% and 0.48% on $AlexNet_Conv(I*M)$ and $AlexNet_Conv(I*S*M)$, respectively. It refers that the transfer loss certainly has an impact on improving the performance of classification but the first four layers in AlexNet_Conv scenarios are enough to extract the general features of OpenSARShip so that the constraint of transfer loss would be better not to affect the off-the-shelf layers to restrict the good feature representation. This also verifies the advantage of analyzing the transferability of layers to distinguish the off-the-shelf and adaptation layers.

2) H_Net: In the H_Net scenario, the features of OpenSARShip in layers 4 and 5 are more likely to share the similar distribution with MSTAR than AlexNet_Conv scenarios but the performance of either transferring and fixing the first four layers or fine-tuning all layers is worse, as given in Tables IV and V. The underlying reason lies in the fact that the base network of $AlexNet(I)$ has a stronger ability to extract generic features than $H_Net(S)$ and $H_Net(M)$. Consequently, in lower layers, the transitive transferring via multisource makes an effort of generalizing the features to OpenSAR to improve the performance of the SAR target recognition task, whereas in the high-level layer, the discrepancy dominates the transferability. In our experiments, the ITL approach improves the performance by 1.15% compared with fine-tuning all layers. With the STL algorithm, considering the decline of feature generalization in higher layers, we set layer 4 and layer 5 as the adaptation layers in scenario $H_Net(M)$ and improve the performance of 2.02%, compared with simply fine-tuning all layers. The multiples of the learning rate in each convolution layer are set to 0.1, 0.1, 0.1, 0.5, 1, and 10 in the classification layer. The results are sensitive to the tradeoff value of λ. In the second step of combining the transfer loss and classification loss, λ must be set to a smaller value to constrain the transfer loss due to the major effect on fine-tuning with the classification loss.

When it comes to $H_Net(S*M)$, we observe in Table V that the performance of STL is not as good as ITL. Fig. 14 shows that the generalization of features in lower layers is

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not as good as layer 5 especially in bottom layers, which indicates that the lower layers have the potential to improve the ability of extracting good features by fine-tuning rather than treated as the off-the-shelf layers. ITL and STL improve the performance of recognizing the OpenSARShip by 1.44% and 0.87% in $H_Net(S*M)$, respectively. As a result, combining the transfer loss with the classification loss to fine-tune all layers as the ITL approach is a better choice in $H_Net(S*M)$.

V. Conclusion

In this article, we elaborately explore what network and source tasks are better to transfer, in which layer the features are more generic to transfer and how to effectively transfer in SAR target recognition. We find that the transferability is up to the generalization capacity of the network and the distance between source and target tasks. A small network is appropriate to train with limited labeled SAR targets but when transferring to other SAR target recognition tasks, the feature generality is not enough to extract a good representation. As a result, a larger network trained with a large-scale data set and a source task similar to SAR target recognition are both required. If possible, a deep network pretrained with a large-scale annotated SAR scene data set is a good source to transfer and we have released the resource in [19]. Otherwise, the unlimited unlabeled SAR images are also helpful especially using transitive transfer proposed in this article to transfer knowledge from large-scale data set to small-scale one, with closer distance to the SAR target recognition task. We do not suggest using natural images pretrained model straightforwardly to SAR targets due to the large difference between them, which may result in many specific features in higher layers. Instead, the midlevel features specific to natural images can be generalized to SAR target by transitive transfer with SAR-related tasks. In order to decrease the discrepancy between source and target domains in very high layer, the proposed MK-MMD-based transfer method to separately train the adaptation layer and slightly update the off-the-shelf layers is recommended, which improves the performance than simply fine-tuning all layers in SAR target recognition transferring.

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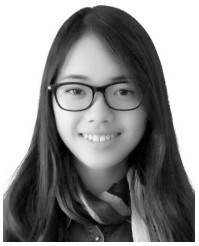
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